



ECE3025: Shielding Effect of MRIs

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Overview

- Magnetic Resonance Imaging (MRI)
 - Strong magnet aligns protons in the body
 - RF coils excite and detect proton signals
 - Gradient coils use spatial encoding to form photon images
- Shielding on MRI machines prevents the leakage of magnetic fields
 - Leakage of magnetic fields interfere with nearby electronic equipment and communication systems
- Shielding protects the MRI itself from external electromagnetic noise that can degrade image quality

Plan of Approach

- Measure the strength of the electromagnetic fields emitted from the MRI machine, specifically focusing on the magnetic field emissions
 - Use a gauss meter to measure the field strength
- Data will be collected at various distances away from the bore opening of the MRI machine
- Magnetic field ranges from the mT range up to about 1T, which captures the fringe field around the 7T MRI where shielding actually matters
- Goal: Evaluate how the electromagnetic field intensity decreases with distance from the MRI machine, as well as assess the effectiveness of the shielding in containing these emissions.

Disclosure

Experiment Assumptions and Conditions

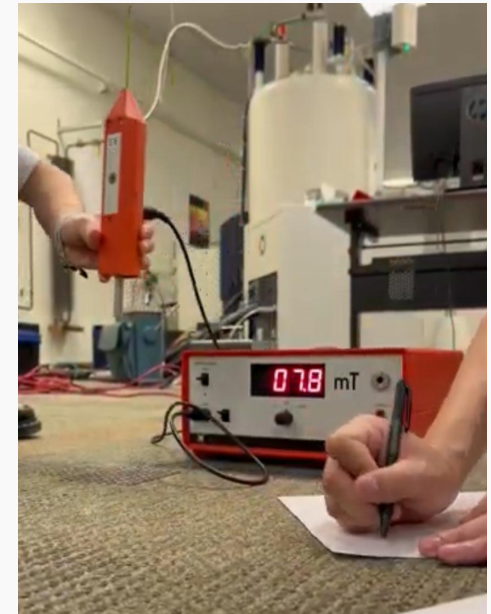
- Room for the animal MRI was completely shielded
- MRI used for experiment was intended for smaller mammals and animals
 - MRI used in experiment is 7T (higher magnetic field means better signal-to-noise ratio (SNR) which means sharper imaging and finer details needed for smaller creatures like animals)
 - MRI typically used for humans ranges from 1.5 to 3T

Experiment Results

Vertical MRI

		Horizontal Distance			
		0.43m	0.60m	1m	1.5m
Vertical Distance	1m	10.0mT	9.9mT	6.1mT	1.5mT
	0.8m	12.9mT	14.1mT	6.5mT	-2.2mT
	0.6m	17.5mT	22.7mT	7.5mT	-9.5mT
	0.4m	23.5mT	48.2mT	11.2mT	-42.2mT
	0.2m	27.0mT	95.3mT	*	*
	0m	51.0mT	120.0mT	*	*

* no data collected as the MRI machine and surrounding equipment blocked the meter



Horizontal Bruker PharmaScan 7T MRI Results

Box Gauss Meter



Distance (m)	Magnetic Field (mT)
1	5.2
0.8	5.0
0.6	4.5
0.4	1.8
0.2	-3.3
0	235

F.W. Bell 5200 Series Gaussmeter

Distance (m)	Magnetic Field (mT)
1	-32.9
0.8	-34.8
0.6	-40.1
0.4	-55.4
0.2	-136.5
0	-612



F.W. Bell 5200 Series Gaussmeter (with Glass Transverse Probe)

Distance (m)	Magnetic Field (mT)
1	-30.5
0.8	-30.8
0.6	-31.2
0.4	-32.7
0.2	-53.3
0	182

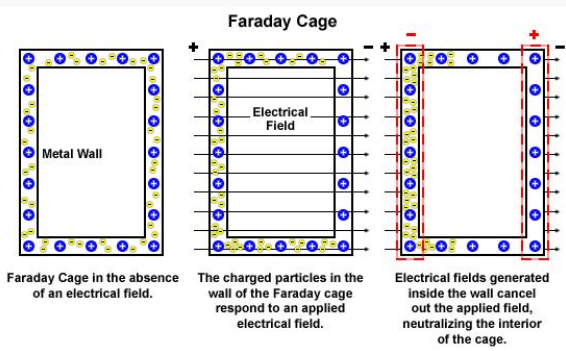


Connection to Class Material

Faraday's Law (Differential Form):

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

- Explains the point to point basis of how external interference is dealt with in the current scope as the room was lined with copper to incorporate RF shielding.
- Walkthrough of the process
 1. The copper makes it so that interference from outside the room (such as a radio wave with its own changing magnetic field) would create a current from the resulting electric field.
 2. That current would induce a magnetic field that would cancel out the initial magnetic from the external source, hence shielding.
- This understanding of Faraday's Law helps us to understand the physical implementation of Faraday's Cage



Connection to Class Material

Ampere's Law (Integral Form)

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

- The coils within the MRI machine create measurable magnetic fields described by Ampere's law
- Ideally, we would have been able to measure the current within the coils of the MRI machine

Superposition Principle

$$\vec{B}(r) = \sum_{i=1}^N \vec{B}_i(r)$$

- The current paths within the MRI machine is set up as a [solenoid](#), which effectively contains several loops side by side.
- The property of superposition allows us to sum up the magnetic field contribution from each individual ring into the total contribution as the solenoid as a whole, rather than needing to derive a new method for the entire object.

Connection to Class Material

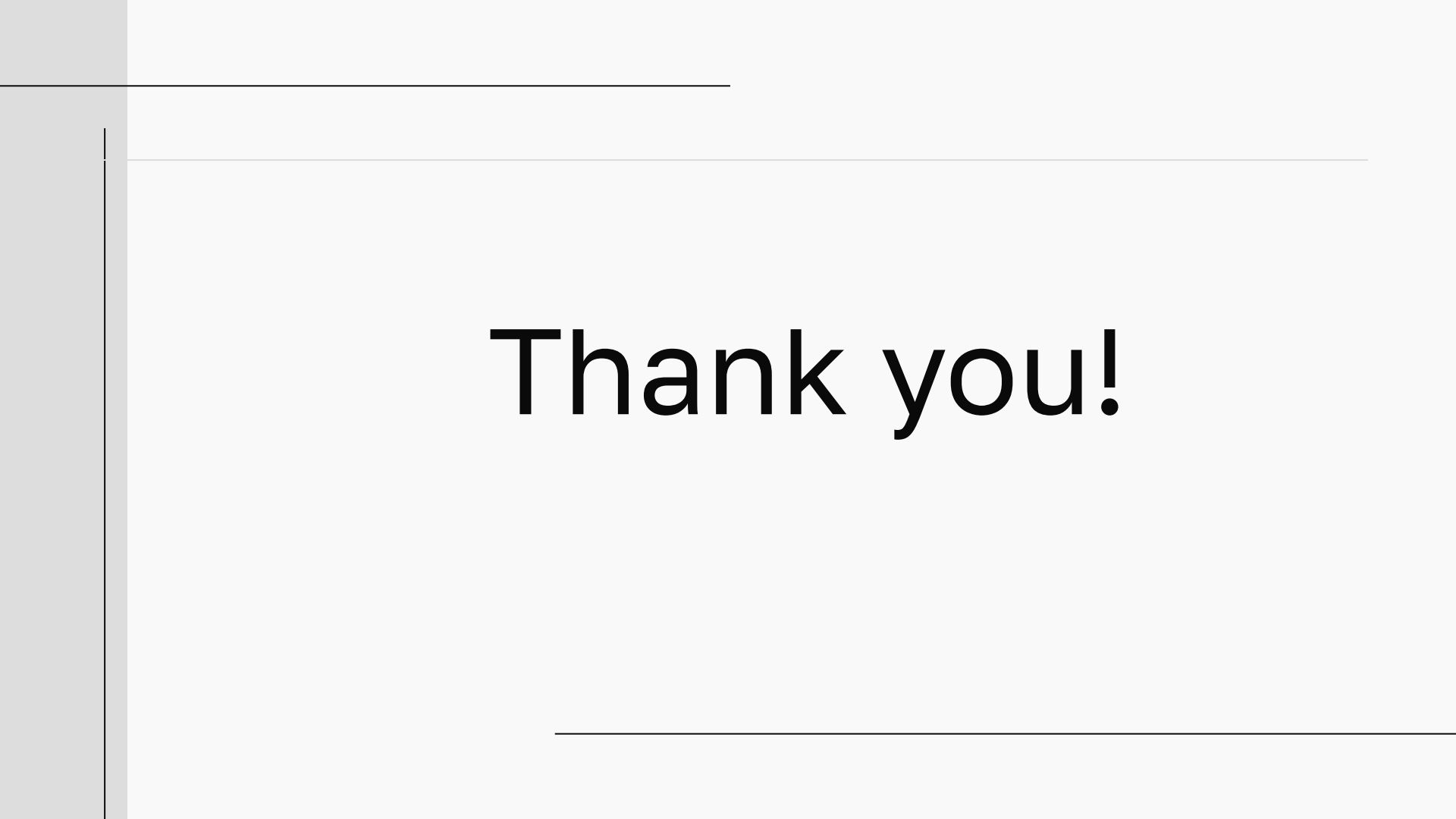
Central Field Uniformity

$$|\vec{B}| \approx \mu_0 n I$$

- Once we started approaching the MRI, the readings began approaching a near-constant value
- This phenomenon is due to the physics of a solenoid, which, when worked out, yields the equation to the left.

Post Experiment Reflection

- Use stricter and more controlled testing conditions
- Develop a clear, well-defined testing plan before data collection
- Ensure gaussmeters are calibrated or account for measurement offsets
- Consistency in probe positioning and orientation
 - As there could be human error in distance/position measurement
- External magnetic interference
 - ex: Earth's Magnetic Field, nearby electronics

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Thank you!

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Abstract:

https://docs.google.com/document/d/1uzmhUPHkGaabNw5WOH30CvMAa_dZIGIW7G5TU5ycSQc/edit?usp=sharing

Yall can decide whatever theme looks good

Results:

https://drive.google.com/file/d/13DiVX-l8jdhmAbgd4TeUGVt0_cmPWWWH/view?usp=sharing

Video: https://drive.google.com/file/d/1l1J3P-vsaTPlgr_LOGmmK8MSxkiGimlj/view?usp=sharing

Datasheets:

<https://drive.google.com/file/d/1BfEYBtGMbAFuVSnzHIXQq1akXgCfHBvA/view?usp=sharing>